

Derivatives and Fixed Income PGE M1 Fall 2026

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Session 1: Time Value of Money & Forwards

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Arrive on time.

Don't cheat Edusign (automatic F). I will count # in classroom.

Electronic devices. No laptops. Tablets are allowed only for note taking.

General principle. Don't be a distraction.

About this class

Stop me if you need me to repeat.

Ask questions if you don't understand.

Participate in class discussions, and be respectful towards others.

Consider the following problem

$$€100 + \$100 = ?$$

Cannot add currencies without first converting into common currency

$$€100 + \$100 \times (0.86€/\$) = €186$$

$$€100 \times (\$1.16/€) + \$100 = \$216$$

A euro today is not the same as a euro in one year.

- ▷ The discount rate is an exchange rate that equalizes the value of money at different points in time.
- ▷ Finance is the valuation of cash flows that accrue in the future.

Today, we are going to use r to denote the discount rate / interest rate.

For now, we are going to ignore risk entirely.

Suppose the interest rate is 5%. What is the value of €100 in one year if you deposit it in a bank?

For a single cash flow:

$$\text{Future Value (FV)} = C \times (1 + r)^t$$

where

- ▷ C = cash flow
- ▷ r = discount rate
- ▷ t = time

Suppose the interest rate is 5%. You need €105 in one year.
How much money do you deposit in a bank **today** to make sure you have €105 one year from now?

For a single cash flow:

$$\text{Present Value (PV)} = \frac{C}{(1 + r)^t}$$

Searching for r

What is the discount rate that makes you indifferent between receiving €100 today versus €110 in one year?

Suppose you deposit €100 in a bank today at an annual rate of 5%. However, interest compounds **monthly**.

- a. What is the value in your bank account after 1 year?

Formula:

$$\begin{aligned}FV &= PV \times (1 + r/m)^{mT} \\&= 100 \times (1 + .05/12)^{12} \\&= 105.1\end{aligned}$$

Suppose you deposit €100 in a bank today at an annual rate of 5%. However, interest compounds **monthly**.

b. What is the effective annual rate?

Formula:

$$\begin{aligned} EAR &= (1 + r/m)^{mT} - 1 \\ &= (1 + .05/12)^{12} - 1 = .051 \end{aligned}$$

Continuous Compounding

Suppose r is the continuously compounded rate.

Definition. an obligation to buy or sell an asset at a fixed time in the future for a specified price agreed upon **today**.

Example. you agree to purchase 350 barrels of oil at €100 per barrel in one year.

underlying asset =

expiration date =

forward (delivery) price =

Forward contract to exchange barrel of oil at €100 in 1 year

What happens TODAY?

Buyer:

1. **Commits** to buy a barrel of oil at €100 in a year
2. Holds **LONG** position

Seller:

1. **Commits** to sell a barrel of oil at €100 in a year
2. Holds **SHORT** position

No money changes hands!!

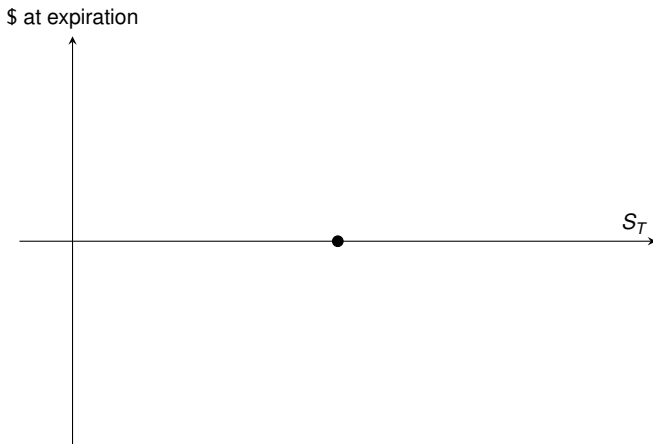
- ▷ Money changes hands only on the day of settlement.
- ▷ Typically negotiated between two private parties in OTC markets.
 - (+) Customizable
 - (-) Hard to leave position before expiry
 - (-) Counterparty risk

What can be hedged with forwards?

- ▷ Interest rate risk
- ▷ Currency risk
- ▷ Stocks
- ▷ Energy (crude oil, natural gas)
- ▷ Metals (gold, copper)
- ▷ Grains (wheat, corn, soybean)
- ▷ Livestock (pork, turkey)

Payoff Diagram: Long Forward

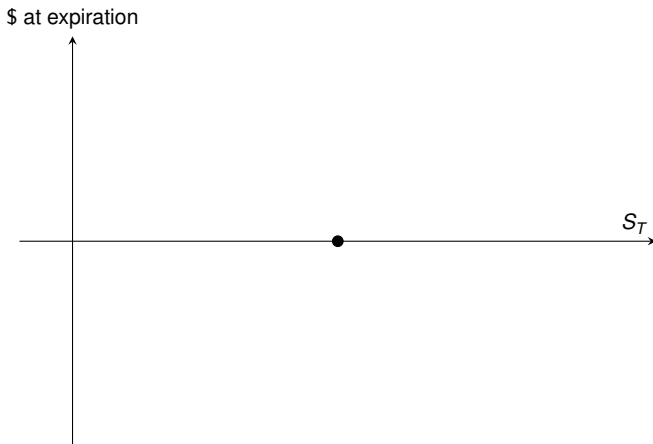
Suppose Taylor Swift is scheduled for a concert in Paris next year. Tickets are sold-out and must be purchased in secondary markets (Vivid Seats, StubHub, etc). You enter a long forward position for delivery of 1 ticket next year at price F . Draw the payoff diagram.



Payoff Table: Long Forward

Payoff Diagram: Short Forward

Suppose Taylor Swift is scheduled for a concert in Paris next year. Tickets are sold-out and must be purchased in secondary markets (Vivid Seats, StubHub, etc). You enter a **short** forward position for delivery of 1 ticket next year at price F . Draw the payoff diagram.



Payoff Table: Short Forward

Payoff Table: Other actions

	Today	Payoff at T
Long Stock	$-S_0$	S_T
Short Stock	$+S_0$	$-S_T$
Borrow (sell bond)	$+S_0$	$-S_0(1+r)^T$
Lend (buy bond)	$-S_0$	$S_0(1+r)^T$

- Technically, you can borrow and lend any amount today but S_0 is common.

Suppose a 1-year forward contract on Apple has a delivery price of \$326.5. The current stock price is \$330. The annual riskless rate is 5%. Show how you could make arbitrage profits.

Assume the “fair value” of a forward is

$$F = S_0 \times (1 + r)^T$$

where S_0 is initial value of the stock price

$$F = 330 \times 1.05 = 346.5$$

Arbitrage with Forwards

Suppose a 1-year forward contract on Apple has a delivery price of \$326.5. The current stock price is \$330. The annual riskless rate is 5%. Show how you could make arbitrage profits.

Arbitrage with Forwards

What if the forward price is \$366.5 (instead of 326.5)? The annual riskless rate is 5%. Show how you could make arbitrage profits.

Forward Pricing

LOOP. If two portfolios have the same payoffs, by no-arbitrage, they **MUST** have the same price.

Example

The price of gold is currently \$1,200 per ounce. The forward price for delivery in 1 year is \$1,300 per ounce. An arbitrageur can borrow money at 3% per annum. What should the arbitrageur do? Assume that the cost of storing gold is zero and that gold provides no income.

Example

An investor enters into a short forward contract to sell 100,000 British pounds for U.S. dollars at an exchange rate of 1.3000 USD per pound. How much does the investor gain or lose if the exchange rate at the end of the contract is (a) 1.2900 and (b) 1.3200?